

MGroups: A Mathematica Package for Finite Group Theory

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Summary

Group theory plays fundamental roles in both pure and applied mathematics, serving as a powerful framework for studying symmetry, structure, and transformations. Its applications span diverse areas such as algebra, number theory, geometry, physics, cryptography, and computer science among others. In recent years, computational tools have become increasingly important for exploring and experimenting with abstract algebraic structures. **MGroups** is a Mathematica package developed to support basic study of finite group structures. It offers a user-friendly interface for performing a wide range of computations involving finite groups such as cyclic groups, unit groups modulo n , dihedral groups, symmetric groups, and more. The package allows users to construct and analyse groups in terms of Cayley tables/graphs, subgroup enumeration and lattices, group homomorphisms, cosets, and related structural investigations.

Statement of need

Computational tools play a vital role in modern mathematics by automating complex and often tedious calculations, thereby freeing users from manual labour and enabling deeper exploration of abstract concepts. In educational settings, such tools are known to enhance learning by supporting experimentation, visualisation, and active engagement with abstract structures, particularly when used alongside formal theory ([Artigue, 2002](#)). Computational algebra and related mathematics analysis software have been shown to especially support pedagogy by enabling exploration, visualisation, and shifts in classroom practice beyond routine computation ([Pierce & Stacey, 2010](#)). One of the most powerful and well-established systems in discrete algebras is *GAP – Groups, Algorithms, and Programming, Version 4.15.1* ([2025](#)), which provides extensive capabilities for working with groups, rings, vector spaces, and more. GAP is widely used in research and advanced applications, offering deep algorithmic functionality and a rich library of algebraic data. However, its syntax and interface can be less accessible to beginners or those unfamiliar with command-line environments.

In contrast, MGroups is a Mathematica package developed with an emphasis on ease of use, accessibility, and pedagogical support. It is designed primarily for undergraduate students beginning with group theory, as well as for educators seeking a user-friendly tool to demonstrate group-theoretic concepts interactively. The package leverages Mathematica's intuitive syntax and symbolic computation engine to simplify many standard tasks, such as constructing Cayley tables, studying subgroups structures, exploring group morphisms, and so on. These operations can typically be performed using simple, one-line commands, greatly lowering the barrier to entry for new learners.

While tools like GAP remain the appropriate choice for large-scale or research-level computations, packages like MGroups fill a complementary niche by providing a gentle and engaging entry point to computational group theory within familiar and friendly environments.

53 State of the field

54 A variety of computational tools for group theory are currently available, each addressing
 55 different audiences and use cases. Dedicated systems such as GAP provide extensive
 56 algorithmic depth and are widely used in research-level computations involving groups, rings,
 57 and representations. General computer algebra systems (Bosma et al., 1997; Maplesoft,
 58 2025; Meurer et al., 2017; The Sage Developers, 2024; Wolfram Research, 2025) include
 59 group-theoretic functionality as part of a broader symbolic framework, though such functionality
 60 is often dispersed and not tailored specifically to introductory group theory. Research-oriented
 61 systems such as Magma and SageMath emphasise algorithmic depth and completeness, while
 62 general-purpose CAS such as Mathematica and Maple integrate group theory alongside many
 63 other mathematical domains. Within the Mathematica ecosystem, specialised packages
 64 such as GTPack (Geilhufe & Hergert, 2018) and GroupMath (Fonseca, 2021) extend native
 65 capabilities, with a particular emphasis on representation theory, symmetry analysis, and
 66 applications in physics. The Group Explorer (Carter & Ellis, 2019) is an excellent visualisation
 67 tool that uses pre-computed data using GAP and provides interactive representations of
 68 finite groups. Programming-language libraries, such as the group-math (Pillmore & Mullanix,
 69 2021) package in Haskell, support abstract algebraic structures within software development
 70 workflows but are not designed as interactive educational tools. In contrast, MGroups is
 71 designed as a lightweight, pedagogy-oriented Mathematica package that focuses on core
 72 topics in finite group theory, offering a unified and accessible interface aimed at students and
 73 instructors rather than large-scale or research-oriented computation.

74 Conclusion

75 MGroups provides an accessible and practical entry point to computational finite group theory
 76 for Mathematica users. By prioritising clarity, ease of use, and pedagogical relevance, it
 77 complements existing research-oriented software and supports both teaching and exploratory
 78 learning in group theory.

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